

WORKSHEET – SOLUTIONS AND SOLUBILITY

Section A – Multiple Choice Questions ($1 \times 5 = 5$)

1. Which of the following is an example of a gas dissolved in a liquid?

- A. Salt in water
- B. Carbon dioxide in water
- C. Copper in zinc
- D. Alcohol in water

2. A solution in which more solute can still dissolve at a given temperature is called:

- A. Saturated solution
- B. Unsaturated solution
- C. Concentrated solution
- D. Suspension

3. Which factor generally increases the solubility of most solid solutes in water?

- A. Decrease in temperature
- B. Increase in temperature
- C. Decrease in pressure
- D. Stirring only

4. The solubility of a gas in a liquid generally increases with:

- A. Increase in temperature
- B. Decrease in pressure
- C. Increase in pressure
- D. Increase in volume

5. Which of the following is an example of a liquid-in-liquid solution?

- A. Salt in water
- B. Oxygen in water
- C. Alcohol in water
- D. Brass

Section B – Very Short Answer Questions ($2 \times 3 = 6$)

6. Give one example each of: (a) solid in liquid solution, (b) liquid in liquid solution, and (c) gas in liquid solution.

7. Differentiate between a saturated solution and an unsaturated solution on the basis of the amount of solute that can dissolve at a given temperature.

8. State any two factors that affect the solubility of a substance in a solvent.

Section C – Short Answer Questions (3 × 2 = 6)

9. A student adds sugar to water at room temperature. Initially, all the sugar dissolves, but after adding more sugar, some remains undissolved at the bottom. Identify the type of solution before and after the excess sugar remains undissolved. Give a reason.

10. A sealed bottle of a carbonated drink is opened. Bubbles of carbon dioxide escape rapidly. Explain this observation in terms of pressure and solubility of gases.

Section D – Competency-Based Question (3 marks)

A student prepares two sugar solutions using the same quantity of water at the same temperature. In Solution A, more sugar can still dissolve, whereas in Solution B, no more sugar dissolves.

10. Answer the following:

- (a) Identify Solution A and Solution B.
- (b) Explain the difference between the two solutions.
- (c) Suggest one change in conditions that may allow more sugar to dissolve in Solution B.

Quick Revision Box

- A solution is a homogeneous mixture of solute and solvent.
- Examples: salt in water (solid in liquid), alcohol in water (liquid in liquid), carbon dioxide in water (gas in liquid).
- Solubility is affected by factors such as temperature and pressure; the effect depends on the nature of the solute and solvent.
- For gases, solubility generally increases with pressure and decreases with increase in temperature.
- An unsaturated solution can dissolve more solute at a given temperature.
- A saturated solution contains the maximum amount of solute that can dissolve at a given temperature.

Answer Key

1. B

2. B

3. B

4. C

5. C

6. Salt in water; alcohol in water; carbon dioxide in water.

7. Saturated solution: no more solute can dissolve at that temperature; unsaturated solution: more solute can dissolve.

8. Temperature and pressure (appropriate factor depends on the nature of solute; pressure is especially significant for gases).

9. Before excess sugar remains: unsaturated; after undissolved sugar persists: saturated, because the solution has reached its solubility limit at that temperature.

10. (a) A – unsaturated; B – saturated. (b) More solute can dissolve in A, while B has reached its solubility limit. (c) Increase the temperature, for a typical solid such as sugar, to increase its solubility.